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main.c... Output

```cpp
#include <iostream>
#include <iomanip>
using namespace std;
int main() {
    // 1D array holding body temperature (°C) for 6 ICU patients
    double temp[6] = {36.8, 38.2, 37.1, 39.0, 36.5, 37.9};
    int i, j;
    double sum = 0.0, average = 0.0;
    double maxT = temp[0], minT = temp[0];
    double searchValue = 37.1;
    int feverCount = 0;
    double feverThreshold = 37.5;
    double tempSwap;

    // Output formatting setup:
    cout << fixed << setprecision(1);

    // --------- TODO 1: Display all readings ---------
    // Loop through temp[0..5] and print "Patient i: value °C"
    cout << "--- Patient Temperature Readings ---" << endl;
    for (i = 0; i < 6; i++) {
```

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```
    for (i = 0; i < 6; i++) {
        cout << "Patient " << i + 1 << ": " << temp[i] << " C" << endl;
    }
    cout << endl;

    // --------- TODO 2: Find the highest and lowest temperature ---------
    // Loop through the array, updating maxT / minT as needed.
    int maxIndex = 0, minIndex = 0;
    for (i = 1; i < 6; i++) {
      if (temp[i] > maxT) {
     maxT = temp[i];
     maxIndex = i;
        }
      if (temp[i] < minT) {
      minT = temp[i];
     minIndex = i;
        }
    }
    cout << "Maximum = " << maxT << " C (Patient " << maxIndex + 1 << ")" << endl;
    cout << "Minimum = " << m                "
        C (Patient " << minIndex + 1
```

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main.c... | Output

```
    cout << "Minimum = " << minT << " C (Patient " << minIndex + 1 << ")" << endl;
    cout << endl;

    // --------- TODO 3: Compute the sum and average temperature ---------
    // Add every element to sum, then divide by 6 for average.
    for (i = 0; i < 6; i++) {
        sum += temp[i];
    }
    average = sum / 6.0;
    cout << "Sum = " << sum << ", Average = " << setprecision(4) << average << " C" << endl;
    cout << endl;
    cout << fixed << setprecision(1);
    // --------- TODO 4: Search for a patient with a specific temperature ---------
    // Search for searchValue in the array and print match or "No matching patient"
    cout << "Searching for temperature: " << searchValue << " C" << endl;
    bool found = false;
```

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main.c... Output

```
        << " C" << endl;
    bool found = false;
    for (i = 0; i < 6; i++) {
    if (temp[i] == searchValue) {
  cout << "Found: Patient " << i + 1 << " has " << searchValue << " C" << endl;
     found = true;
      break;
        }
    }
    if (!found) {
        cout << "No matching patient" << endl;
    }
    cout << endl;
    // --------- TODO 5: Count feverish patients ---------
    // Loop through the array and count how many readings exceed feverThreshold (37.5°C).
    for (i = 0; i < 6; i++) {
        if (temp[i] > feverThreshold) {
            feverCount++;
        }
    }
```

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main.c... Output

```
cout << "Fever count = " << feverCount << " (readings exceed " << feverThreshold << " C)" << endl;
cout << endl;
// --------- TODO 6: Sort by temperature ---------
// 1. Sort the array ascending using Bubble Sort logic
for (i = 0; i < 5; i++) {
for (j = 0; j < 5 - i; j++) {
if (temp[j] > temp[j + 1]) {
tempSwap = temp[j];
 temp[j] = temp[j + 1];
temp[j + 1] = tempSwap;
        }
    }
}
cout << "Ascending sort: ";
for (i = 0; i < 6; i++) {
    cout << temp[i] << " ";
}
cout << endl;

// 2. Sort the array descending for triage (highest fever first)
for (i = 0; i < 5; i++) {
```

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main.c... Output

```
    for (i = 0; i < 6; i++) {
        cout << temp[i] << " ";
    }
    cout << endl;

    // 2. Sort the array descending for triage (highest fever first)
    for (i = 0; i < 5; i++) {
        for (j = 0; j < 5 - i; j++) {
            if (temp[j] < temp[j + 1]) {
                tempSwap = temp[j];
                temp[j] = temp[j + 1];
                temp[j + 1] = tempSwap;
            }
        }
    }
    cout << "Descending sort: ";
    for (i = 0; i < 6; i++) {
        cout << temp[i] << " ";
    }
    cout << endl;
    return 0;
}
```

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main.c... Output

```
--- Patient Temperature Readings ---
Patient 1: 36.8 C
Patient 2: 38.2 C
Patient 3: 37.1 C
Patient 4: 39.0 C
Patient 5: 36.5 C
Patient 6: 37.9 C

Maximum = 39.0 C (Patient 4)
Minimum = 36.5 C (Patient 5)

Sum = 225.5, Average = 37.5833 C

Searching for temperature: 37.1 C
Found: Patient 3 has 37.1 C

Fever count = 3 (readings exceed 37.5 C)

Ascending sort: 36.5 36.8 37.1 37.9 38.2 39
    .0
Descending sort: 39.0 38.2 37.9 37.1 36.8 36
    .5


=== Code Execution Successful ===
```